



5.9 Differential equations and Maclaurin series (HL)

Calculus · IB Diploma · Concept guide and original exam-style practice

Understand the concept

A separable first-order differential equation $dy/dx=f(x)g(y)$ is solved by separating variables: $\int 1/g(y) dy = \int f(x) dx$, then integrating both sides and applying any initial condition to find the constant. A Maclaurin series expresses a function as an infinite polynomial expansion about $x=0$: $f(x) = f(0) + f'(0)x + f''(0)x^2/2! + f'''(0)x^3/3! + \dots$, built from successive derivatives evaluated at $x=0$.

Worked example: follow the reasoning

Solve the differential equation $dy/dx = xy$, given $y=2$ when $x=0$.

Separate: $(1/y)dy = x dx$.

Integrate: $\ln|y| = x^2/2 + C$.

$y = e^{(x^2/2+C)} = Ae^{(x^2/2)}$.

Using $y(0)=2$: $A=2$.

Solution: $y = 2e^{(x^2/2)}$.

Quick check · multiple choice

The Maclaurin series for $f(x)=e^x$ begins:

A $1+x+x^2+x^3$

B $1+x+x^2/2+x^3/6$

C $x+x^2/2+x^3/3$

D $1-x+x^2/2-x^3/6$

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 (a) Find the general solution of the differential equation $dy/dx = (y+1)\cos(x)$, separating variables. [9]
- (b) Given that $y=0$ when $x=0$, find the particular solution.
- (c) Find the Maclaurin series for $f(x)=\ln(1+x)$ up to and including the x^3 term.

Final answer: _____



Answers and explanation

Question 1 [9]

(a) Separate: $\frac{1}{y+1} dy = \cos(x) dx$. Integrate: $\ln|y+1| = \sin(x) + C$. So $y+1 = Ae^{(\sin x)}$, $y = Ae^{(\sin x)} - 1$.

(b) $y(0)=0$: $0=Ae^0-1=A-1$, so $A=1$. Particular solution: $y = e^{(\sin x)} - 1$.

(c) $f(x)=\ln(1+x)$: $f(0)=0$. $f'(x)=1/(1+x)$, $f'(0)=1$. $f''(x)=-1/(1+x)^2$, $f''(0)=-1$. $f'''(x)=2/(1+x)^3$, $f'''(0)=2$.

Series: $f(x) \approx 0 + 1(x) + (-1)x^2/2! + 2x^3/3! = x - x^2/2 + x^3/3$.

Quick check answer

B. $f(0)=1$, $f'(0)=1$, $f''(0)=1$, $f'''(0)=1$, so the series is $1+x+x^2/2!+x^3/3!=1+x+x^2/2+x^3/6$.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Mathematics AA subject guide.

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